Reproducible Synthetic Denoising: A PhD Lab QA Experiment

QA Fixture

Abstract—This technical report validates a synthetic CPU workflow, not a new empirical scientific result.

Index Terms—synthetic data, denoising, reproducibility, quality assurance

I. Methodology

We use 128 sine samples with Gaussian noise sigma 0.25 and a three-point moving average. The authored fixture is cited as [1].

II. Experiments and Results

The measured noisy MSE is 0.052396280822; filtered MSE is 0.018616959299. Fresh subprocess replay and independent recomputation agree within 10^{-12}.

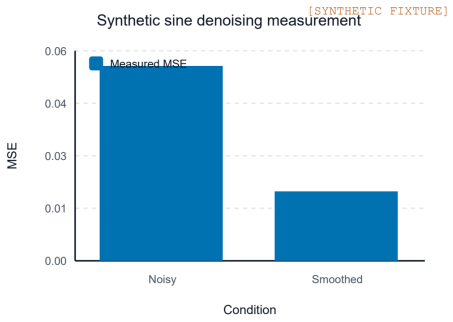


Fig. 1. Measured MSE on the synthetic CPU fixture.

III. Discussion

The dataset is synthetic. No generalization, novelty, provider parity or peer-review approval is claimed.

References

[1] Q. Fixture, “Synthetic signal fixture.”